Module 2: Numerical Measures — Practice

Lesson 3 | Calculate, compare, and interpret

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Preparation

Read Lesson 3. Use the Audit and Restaurant files in the CSV bundle. Submit a concise Quarto report with reproducible calculations, units, and interpretations.

Use the book's **type-6 percentile convention** throughout. Preserve the original CSV values. If an exercise changes an observation, create a separate hypothetical copy and label it clearly.

```
# library(): loads a package so its functions can be used in this session.
# read.csv() imports a CSV into a data frame with named columns.
# The relative path starts at the course project folder; the source CSV is not modified.
library(ggplot2)
audit <- read.csv("data/module02/audit_times.csv")
restaurants <- read.csv("data/module02/restaurant_ratings.csv")
x <- audit$audit_days
```

1. Describe the center

1. Calculate the number of audits, total recorded duration, mean, and median.
2. Show which sorted positions determine the median.
3. Identify the statistical mode using value frequencies. Explain why `mode(x)` does not answer that question in R.
4. Write a sentence explaining the mean and median to an audit manager. Explain why neither number guarantees a future completion date.

2. Quantify variability

1. Calculate minimum, maximum, and range.
2. Calculate the sum of squared deviations from the sample mean, then divide by `n - 1` to obtain the sample variance. Check against `var(x)`.
3. Take its square root and check against `sd(x)`.
4. Give the units of the range, variance, and standard deviation.
5. Explain why dividing by n would produce a different descriptive quantity and why having all 20 CSV rows does not automatically make them the full population of interest.

3. Quartiles and a boxplot

1. Calculate Q_1, median, and Q_3 with `quantile(x, c(.25,.5,.75), type = 6)`.
2. Calculate IQR and the five-number summary.
3. Calculate both 1.5-IQR fences and identify any flagged values.
4. Identify the actual whisker endpoints. Explain how they differ from the fences and the overall maximum.
5. Draw a boxplot with those quartiles and whiskers. Label its units and state the convention. If using a software default, check that its quartiles match your stated method for these data.
6. State what you would investigate before deciding whether a flagged observation is erroneous.

4. Sensitivity and relative measures

1. Replace only the largest audit duration with **60 days in a separate copy**.
2. Compare mean, median, sample SD, range, and IQR before and after the hypothetical change.
3. Explain which measures changed most and why. Do not treat the hypothetical dataset as a corrected source file.
4. Calculate the CV for the original audits and interpret it.
5. Calculate the sample z-score of the original 33-day audit. Explain why it may be flagged by the IQR rule while having a z-score smaller than 3.
6. Explain why a CV for Celsius temperature is not meaningful in the same way as a CV for elapsed duration.

5. Return to the restaurant question

For each quality category, calculate group size, mean, median, sample SD, Q_1, Q_3, and IQR. Preserve the order Good, Very Good, Excellent.

1. Which group has the highest median price? Which has the smallest IQR?
2. Calculate the difference between the Excellent and Good median prices, in USD. Describe it as a difference in group medians, not as an individual restaurant's price change.
3. Compare the groups with side-by-side boxplots, using the same quartile convention as the table.
4. Write a short interpretation of center, spread, and overlap. Explain why these data do not prove that changing a rating causes the price to change.

6. Explain the limitations of a summary

Use the teaching-comparisons CSV, separate from the textbook bundle.

1. For the seven dollar values (`example == "extreme_value"`), calculate the mean and median. Which better describes the middle of the observations? When would the mean still be useful? Explain what information is lost when these seven observations are compressed into a single mean. Does the mode answer the same question as the median?
2. For S1–S5 (`example == "same_mean"`), verify that every mean is 200 and calculate each range. Explain what the mean alone conceals.
3. For A and B (`example == "same_range"`), calculate mean, range, mean absolute deviation about the mean, and sample SD. Explain how two sets with identical means and ranges can differ internally.
4. For the seven dollar amounts, calculate the range and explain why USD 99 does not describe the spread of most amounts. If only USD 100 became USD 1000 in a hypothetical copy, what would the range be? What would remain unchanged about the other six amounts?
5. Explain why replacing range with SD does not remove sensitivity to extreme values. What additional information do median, IQR, and a plot provide?

Reporting check

Every table and chart should make the units and group sizes clear. Use English for statistical explanations. Arabic narration is reserved for introducing business cases; it is not required for your calculation answers.

Return to Lesson 3.